

Tapas Ranjan Rath

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Research Profile

Computational cognitive scientist studying context effects, bounded rationality, and human–AI interaction. Methods span behavioral experiments, Bayesian and mixed-effects modeling, agent-based simulation, and reinforcement-learning benchmarks. Current extension: AI alignment questions on sense of agency, value conflict, cooperation, norm formation, and multi-agent systems aligned to plural human values.

Education

Indian Institute of Technology Kanpur

Ph.D., Cognitive Science

2019–2026

Thesis submitted: 8 Jul 2026

- Thesis: *When and Why Context Matters: Establishing Boundary Conditions and Mechanisms of the Attraction Effect in Decision-Making*
- Supervisors: Prof. Narayanan Srinivasan and Prof. Nisheeth Srivastava

SRM University, Chennai

B.Tech., Electronics and Instrumentation Engineering

2008–2012

Research Experience

Ph.D. Researcher, Cognitive Science

Indian Institute of Technology Kanpur

2019–2026

Kanpur, India

- Designed behavioral experiments in PsychoPy and JavaScript to study attraction effects, pairwise comparison, subjective equivalence, and trade-off difficulty in multi-attribute choice.
- Built simulation frameworks and Bayesian/mixed-effects models to evaluate the robustness of standard decision metrics under heterogeneous preferences.
- Implemented reinforcement-learning agents, including DRQN/POMDP models in PyTorch, to benchmark human adaptive decision-making and cognitive flexibility.
- Developing the KARMA architecture and computational testbed for proxy agency, plural-values alignment, and role-invariant representation interventions in multi-agent reinforcement learning; commons ecology calibrated and pilot interventions underway.

Publications

Journal Articles

- Rath, T., Srinivasan, N., & Srivastava, N. (2025). **The attraction effect in perceptual decision-making: A case of dominance asymmetry.** *Frontiers in Psychology*. [Link]

Conference Papers

- Rath, T., Srivastava, N., & Srinivasan, N. (2026). **A self-directed expanded judgment paradigm: Isolating the pairwise mechanism of the attraction effect.** Presented orally at CogSci 2026 in July (140/1511; 9.3%); *Proceedings of the 48th Annual Meeting of the Cognitive Science Society*. [Link]
- Rath, T., & Marupudi, V. (2025). **Re-evaluating the numerical-perceptual distinction in the attraction effect.** *Proceedings of the 47th Annual Meeting of the Cognitive Science Society (CogSci 2025)*. [Link]

Manuscripts in Preparation

- Rath, T., Srivastava, N., & Srinivasan, N. (Submitted to the Journal of Mathematical Psychology). **Unmasking the flaws of triplet-triplet attraction effect measures: Via mathematical analysis and agent-based simulations.** PsyArXiv preprint available. [DOI]
- Rath, T., Srivastava, N., & Srinivasan, N. (in preparation). **Pairwise comparison architecture in context-dependent choice.** Target journal: *Cognitive Science*.
- Rath, T., Srivastava, N., & Srinivasan, N. (in preparation). **When difficult trade-offs make decoys work: Subjective equivalence geometry in the attraction effect.** Target journal: *Judgment and Decision Making*.
- Rath, T. (in preparation). **When decoys are easy to reject: Comparative asymmetry and context effects in consumer product-pack choice.** Target journal: *Journal of Consumer Psychology* (Short Reports).
- Rath, T. (in preparation). **The Proxy Agency Moral Shield: Extended Self, Semantic Fluency, and the Ethics of Agentic AI.** Target journal: *Minds and Machines*.

Awards and Honors

- **Student Travel Grant, Cognitive Science Society (CogSci 2026)** – \$1,500 USD. Awarded to a first author on a highly rated paper accepted for oral presentation.
- **Psychonomic Society Student Travel Award from Developing Nations (2026)** – \$2,000 USD. Awarded for the abstract *Deliberation Eliminates Decoys Rather Than Amplifying Attraction*.
- **PhD Institute Assistantship, IIT Kanpur (MoE/MHRD)** (2019–2026) – Ministry of Education doctoral study support.

Selected Talks and Presentations

- Rath, T. (2026). **Deliberation eliminates decoys rather than amplifying attraction**. Poster to be presented at the 67th Annual Meeting of the Psychonomic Society, San Diego.
- Rath, T., Srivastava, N., & Srinivasan, N. (2024). **Perceptual stimuli with difficult-to-trade-off attribute values show a positive attraction effect**. Poster presented at the Society for Judgment and Decision Making Annual Meeting, New York City.
- Rath, T., Srivastava, N., & Srinivasan, N. (2024). **De-obfuscating context effects in decision-making: A simulation study**. Paper presented at Virtual MathPsych/ICCM 2024. [Link]
- Rath, T., Srivastava, N., & Srinivasan, N. (2024). **Attribute trade-off difficulty modulates the asymmetric-dominance of the decoy**. Presented at the Annual Conference of Cognitive Science, IIT Bombay.

Technical Skills

- **Statistics and modeling:** R (lme4, brms, BayesFactor, ggplot2); Python (Pandas, NumPy, SciPy, Statsmodels, Seaborn, Matplotlib); Bayesian modeling; mixed-effects models; agent-based simulation.
- **AI and computational methods:** PyTorch, reinforcement learning, DRQN, POMDPs, multi-agent simulation, computational modeling of decision processes.
- **Experimental methods:** PsychoPy, OpenSesame, JavaScript, online/offline behavioral experiments, psychophysics, eye tracking with Eyelink 1000 Plus.

Selected Research Code

- **Situated Agency Alignment (KARMA):** Public MARL codebase for proxy agency, plural-values alignment, commons conflict, and role-invariant representation interventions. [GitHub]
- **Robust Context Metrics:** Agent-based simulations testing attraction-effect measures under heterogeneous preferences. [GitHub]

Teaching and Professional Service

- Teaching Assistant, Human-Centered Computing (CGS616), IIT Kanpur Jan–Jul 2024
- Teaching Assistant, Introduction to Cognitive Science (CGS401), IIT Kanpur Jul–Dec 2023
- Professional memberships: Society for Judgment and Decision Making; Society for Mathematical Psychology; Psychonomic Society; Cognitive Science Society.

Prior Industry and Teaching Experience

- Senior Software Engineer, Accenture, Pune 2012–2014
Led QA strategy for enterprise telecom systems and designed automated testing frameworks.
- Physics Instructor/Teacher, Chaitanya's Academy and DPS Kalyanpur 2014–2017
Designed curriculum and assessments for high-school and competitive-exam physics; mentored 500+ students.

References

- **Prof. Narayanan Srinivasan** – Department of Cognitive Science, Indian Institute of Technology Kanpur. nsrini@iitk.ac.in
- **Prof. Nisheeth Srivastava** – Department of Computer Science and Engineering, Indian Institute of Technology Kanpur. nsrivast@iitk.ac.in